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DEFEND YOUR RESEARCH

Your Use of Pronouns Reveals Your Personality

by James W. Pennebaker

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Defend Your Research

HBR puts some surprising findings to the test



James W. Pennebaker is the chair of psychology at the University of Texas at Austin and the author of *The Secret Life of Pronouns: What Our Words Say About Us* (Bloomsbury Press, 2011).

Your Use of Pronouns Reveals Your Personality

The finding: A person's use of function words—the pronouns, articles, prepositions, conjunctions, and auxiliary verbs that are the connective tissue of language—offers deep insights into his or her honesty, stability, and sense of self.

The research: In the 1990s, James Pennebaker helped develop a computer program that counted and categorized words in texts, differentiating content words, which convey meaning, from function words. After analyzing 400,000 texts—including essays by college students, instant messages between lovers, chat room discussions, and press conference transcripts—he concluded that function words are important keys to someone's psychological state and reveal much more than content words do.

The challenge: Can insignificant words really provide a "window to the soul"? **Professor Pennebaker, defend your research.**

Pennebaker: When we began analyzing people's writing and speech, we didn't expect results like this. For instance, when we analyzed poems by writers who committed suicide versus poems by those who didn't, we thought we'd find more dark and negative content words in the suicides' poetry. We didn't—but we did discover significant differences in the frequency of words like "I." In study after study, we kept finding the same thing. When we analyzed military transcripts, we could tell people's relative ranks based on their speech patterns—and again, it was the pronouns, articles, conjunctions, and other function words that made a difference, not the content words.

HBR: Why are function words so important?

In English there are about 500 function words, and about 150 are really common. Content words—nouns, verbs, adjectives, and most adverbs—convey the guts of communication. They're how we express ideas. Function words help shape and shortcut language. People require social skills to use and understand function words, and they're processed in the brain differently. They are the key to understanding relationships between speakers, objects, and other people. When we analyze people's use of function words, we can get a sense of their emotional state and personality, and their age and social class.

Here's a simple, pronoun-heavy sentence: I don't think I buy it.

Ooh. You just revealed something about yourself in that statement. Why did you say "I don't think I buy it" instead of "I don't buy it" or even "That's ridiculous"? Pronouns tell us where people focus their attention. If someone uses the pronoun "I," it's a sign of self-focus. Say someone asks "What's the weather outside?" You could answer "It's hot" or "I think it's hot." The "I think" may seem insignificant, but it's quite meaningful. It shows you're more focused on yourself. Depressed people use the word "I" much more often than emotionally stable people. People who are lower in status use "I" much more frequently.

Can you tell if someone's lying by their use of function words?

Yes. A person who's lying tends to use "we" more or use sentences without a first-person pronoun at all. Instead of saying "I didn't take your book," a liar might say "That's not the kind of thing that anyone with integrity would do." People who are honest use exclusive words like "but" and "without" and negations such as "no," "none," and "never" much more frequently. We've analyzed transcripts of court testimony, and the differences in speech patterns are really clear.

Function words sound like two-by-fours: They're important but not meaningful in creating the overall architecture.

You might even think of function words as the nails. It seems natural to pay them little regard. If you type a sentence into Google, its algorithms disregard function words, because it's interested in content. But these words convey important subtleties—"a ring" versus "that ring." In foreign languages, function words often convey people's status relative to one another.

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Key Numbers

Out of 100,000 words in the average English speaker's vocabulary, function words account for only about 500, or

0.5%.

55%

of what we speak, hear, and read in typical speech, however, is made up of these function words.

If you listened to a job interview, what would the use of function words tell you?

It's almost impossible to hear the differences naturally, which is why we use transcripts and computer analysis. Take a person who's depressed. "I" might make up 6.5% of his words, versus 4% for a nondepressed person. That's a huge difference statistically, but our ears can't pick it up. But hypothetically, if I were to listen to an interview, I might consider how the candidate talks about their coworkers at their last job. Do they refer to them as "we" or "they"? That gives you a sense of their relationship to the group. And if you want someone who's really decisive in a position, a person who says "It's hot" rather than "I think it's hot" may be a better fit.

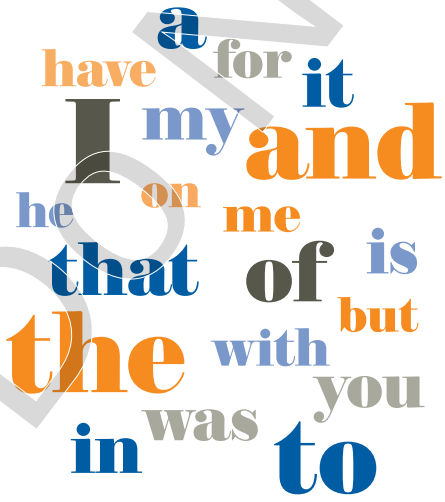
How do people react to your analyses of their speech?

I did it using my own speech and was really surprised. I used the software on everything I wrote—even e-mails. I also developed a recorder that people could wear. It would turn on for 30 seconds every 12 minutes to capture bits of everyday speech. I wore it myself. When I analyzed my speech, it struck me how differently I spoke to my son, who was then 12 years old. With my daughter and my wife, my language was much more informal and personal. With my son it was more cool and detached. I realized I was drawing back from him—I wasn't being psychologically present. This was during a period of some tension in our relationship. He was a typical adolescent and was acting out a bit, and I was responding by being cool and detached, which males stupidly do when we're annoyed. When I realized this, I tried to become more human, emotional, and honest with him.

Are there gender differences in how we use function words?

Most people think men use "I" more, because men are more narcissistic and self-congratulatory. But across studies and cultures, we found that women use "I," "me," and "mine" more. Women are more self-attentive and aware of their internal state. Men use more articles: "a," "an," and "the." That means men talk about objects and things more. You use articles when you're referring to concrete objects, because articles precede concrete nouns. Women also use more third-person pronouns—"he," "she," and "they"—because women talk more about people and relationships, and they're better at managing them. And in many ways, relationships are more complex.

THE 20 MOST COMMONLY USED WORDS



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